



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to PPS FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

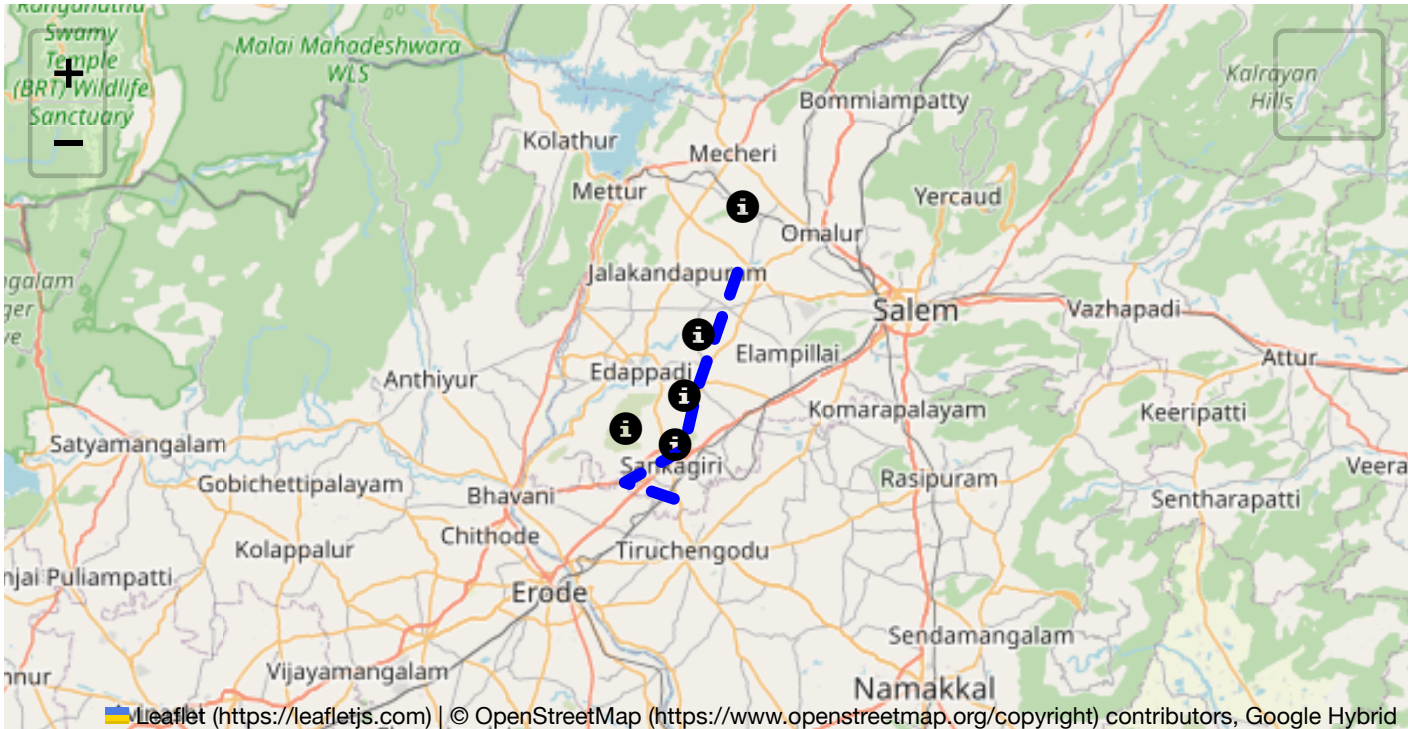
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

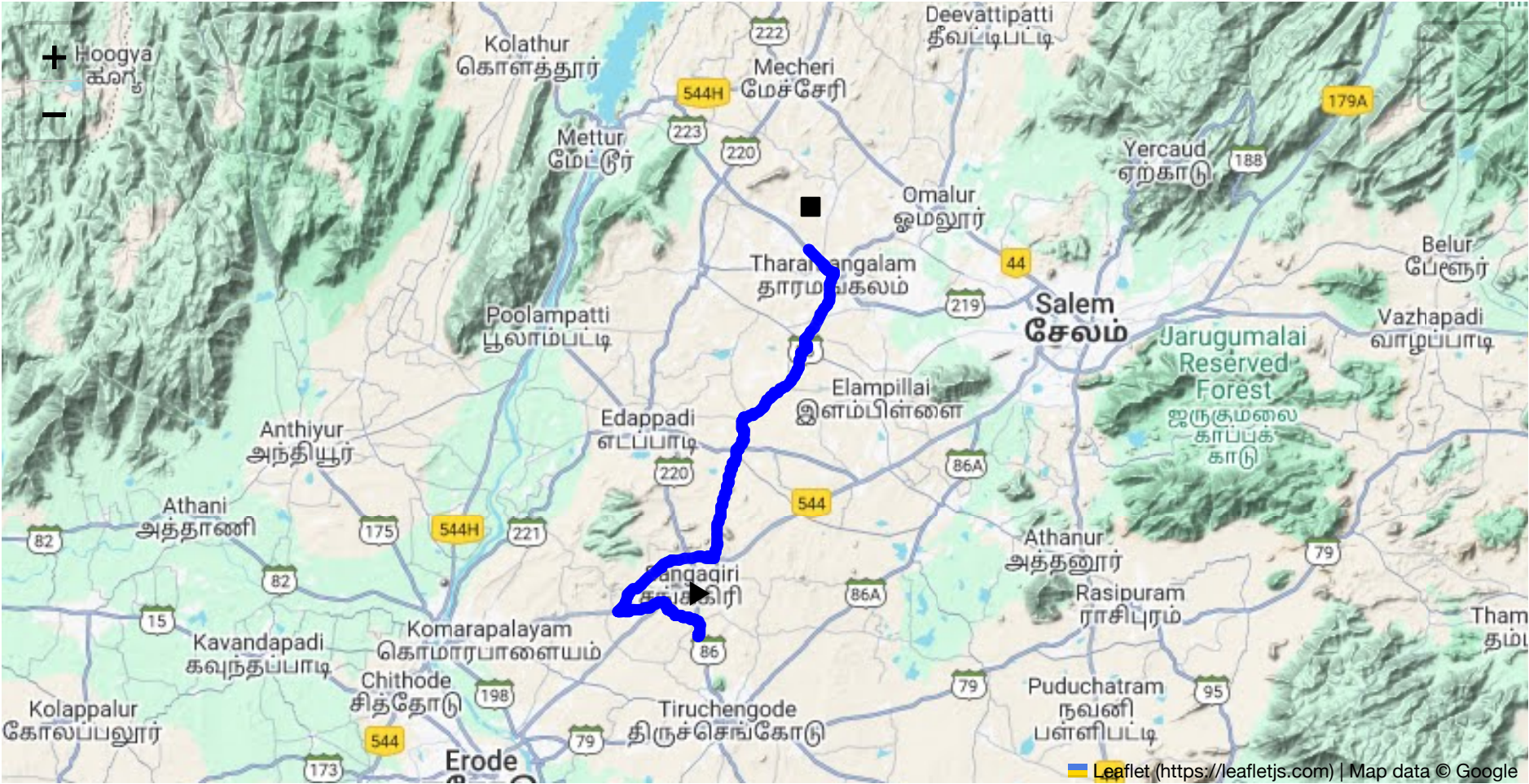
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 46.30 km
Estimated Duration: 1.1 hours
Adjusted Duration (Heavy Vehicle): 1.4 hours

Start: (11.4381, 77.8734)
End: (11.71155396, 77.95330314)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W, Sangagiri, Tamil Nadu to PX63+Q89, Salem, Tamil Nadu passes through Padaiveedu, Ayveli, and Erumaipatti, utilizing Bhavani Main Road and SH 86. The journey covers approximately 46.30 kilometers and is generally completed in around 1.13 hours for heavy vehicles transporting hazardous materials.

Weather Conditions

Typically, Tamil Nadu experiences a tropical climate with dry seasons and heavy rainfall during the monsoon (June to September). The potential weather-related hazards include reduced visibility and road slipperiness due to rain during monsoon, and high temperatures might cause vehicle heating issues during summer months.

Traffic Patterns

- Peak Hours:** Typically, congestion is highest during morning (8 AM - 10 AM) and evening (5 PM - 7 PM) hours due to local commuting.
- Congestion Areas:** Major congestion points likely include town centers like Padaiveedu and intersections on SH 86. During religious or local festivals, traffic can significantly increase.

Road Quality and Infrastructure

- Road Surface:** Generally, the major roads, such as SH 86, are well-maintained, but smaller connecting roads might have potholes or uneven surfaces.
- Signage and Lighting:** Adequate in towns but may be insufficient in rural stretches, requiring caution especially during night-time driving.

Alternative Routes

In emergencies, the following alternatives could be considered:

- Divert via NH 544 if accessible, as it's a better-maintained highway.
- Consider local roads parallel to Bhavani Main Road if traffic conditions allow for it.

Local Regulations on Hazardous Material Transport

Tamil Nadu follows strict guidelines on the transport of hazardous material, including designated routes and times when transport is not allowed typically during peak traffic times for safety reasons.

Historical Incidents

There have been sporadic incidents involving heavy vehicles, particularly due to overloading or mechanical failures, particularly during the monsoon period. However, large-scale incidents involving hazardous materials are rare but necessitate stringent precautionary measures.

Environmental Considerations

- **Sensitive Areas:** Ayveli might have agricultural areas where precautions against leaks and spills are crucial due to proximity to farmlands.
- **Noise and Emissions:** Important to minimize disturbances, especially in smaller towns.

Communication Coverage

Most of the route should have adequate mobile coverage given the proximity to populated areas. However, some stretches between towns might have limited coverage, which can affect emergency communications.

Emergency Response Times

- **Urban Areas (e.g., near Ayveli):** Approx. 15-20 minutes due to proximity to local services.
- **Rural Stretches:** Could experience delays upwards of 30-45 minutes; hence, pre-equipped vehicles with first-aid and emergency response kits are recommended.

Overall Summary of Risk Assessment

The route is moderately safe, with primary concerns involving:

- Weather conditions during monsoon.
- Peak hour traffic leading to possible delivery delays.
- Communication lapses in rural stretches.
- Regulation compliance for the transit of hazardous materials.

Proactive measures such as route familiarization, regular vehicle maintenance checks before transit, and adherence to local regulatory guidelines can mitigate many of these risks effectively. Regular updates on weather and traffic conditions via mobile applications and having contingency plans for emergencies are highly recommended practices.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.45353, 77.85697	30 KM/Hr
5	Turn	Medium	11.45839, 77.85143	30 KM/Hr
6	Turn	Medium	11.46307, 77.84941	30 KM/Hr
7	Turn	Medium	11.46316, 77.84921	30 KM/Hr
8	Turn	High	11.45542, 77.81725	15 KM/Hr
9	Blind Spot	Blind Spot	11.45631, 77.81668	10 KM/Hr
10	Turn	High	11.49409, 77.88004	15 KM/Hr
11	Turn	Medium	11.49420, 77.88016	30 KM/Hr
12	Turn	High	11.49382, 77.88530	15 KM/Hr
13	Turn	High	11.49408, 77.88538	15 KM/Hr
14	Turn	Medium	11.49444, 77.88492	30 KM/Hr
15	Turn	Medium	11.49469, 77.88488	30 KM/Hr
16	Turn	Medium	11.56096, 77.89931	30 KM/Hr
17	Turn	Medium	11.57036, 77.90223	30 KM/Hr
18	Turn	Medium	11.57042, 77.90222	30 KM/Hr
19	Turn	Medium	11.57053, 77.90227	30 KM/Hr
20	Turn	Medium	11.57056, 77.90234	30 KM/Hr
21	Turn	Medium	11.58304, 77.90656	30 KM/Hr
22	Turn	Medium	11.64197, 77.95197	30 KM/Hr
23	Turn	Medium	11.64721, 77.95118	30 KM/Hr
24	Blind Spot	Blind Spot	11.69590, 77.97114	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Dr. Pandiyarajan Hospital	11.692704, 77.969344	30 km/h	Medium
3	clinic	Sun Eye Care	11.6936242, 77.9730948	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
4	hospital	Dr. Sengodan Hospital	11.6959527, 77.9695325	30 km/h	Medium
5	hospital	Veda Maruthuvamanai Hospital	11.6956155, 77.9678122	30 km/h	Medium
6	clinic	E.V.P Vairavel Clinic	11.6954975, 77.9667903	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544

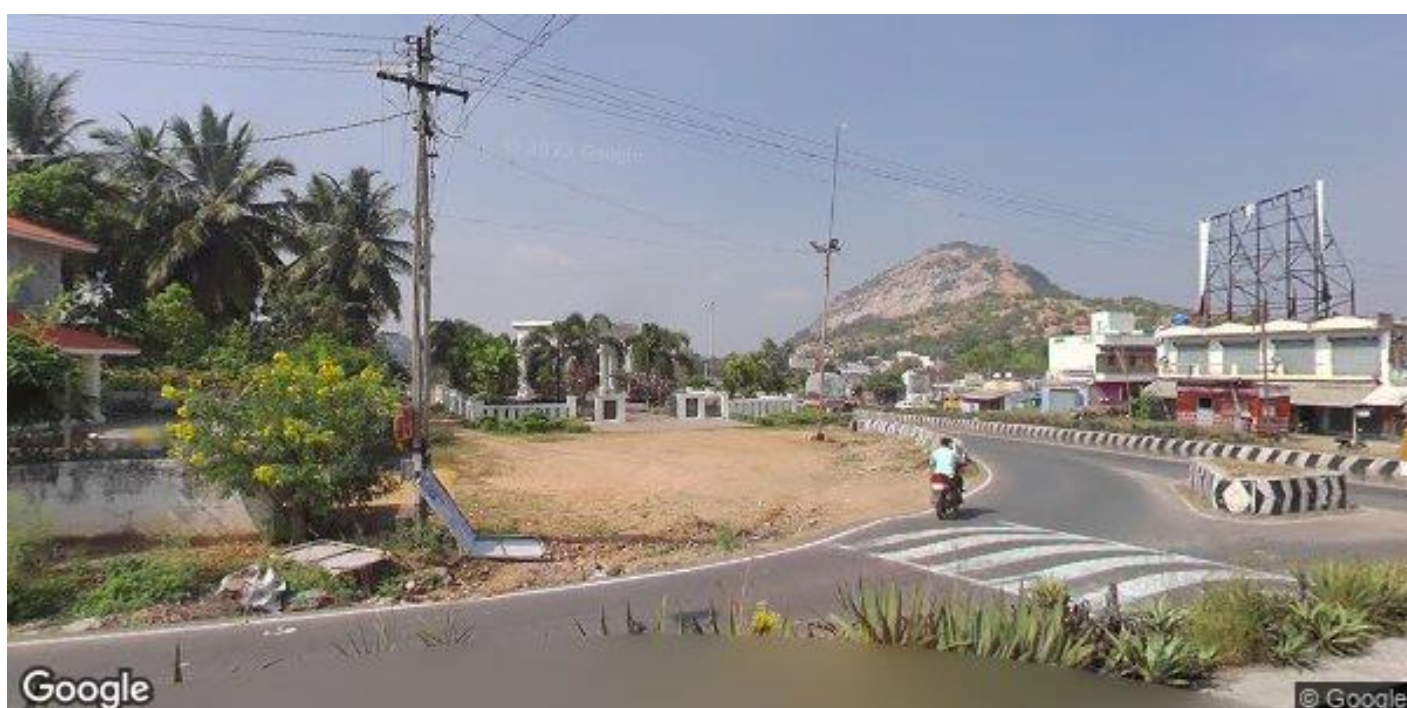


Risk Type: Turn
Risk Level: Medium

Speed Limit: 30 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45839, 77.85143



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46316, 77.84921



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45631, 77.81668



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49409, 77.88004



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49420, 77.88016



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49382, 77.88530



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49408, 77.88538



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49444, 77.88492



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49469, 77.88488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56096, 77.89931



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57036, 77.90223



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57042, 77.90222



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57053, 77.90227



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57056, 77.90234



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58304, 77.90656



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64197, 77.95197



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64721, 77.95118



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.69590, 77.97114
